Akash Dubey

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Education

Rutgers University Honors College, School of Arts and Sciences — New Brunswick, NJ

Bachelor of Science in Computer Science and Mathematics — *Anticipated May 2028*

- **GPA**: 3.95/4.0 | SAT: 1580 | Dean's List. **Coursework**: Systems Programming, Computer Architecture, Algorithms, Data Structures, Stochastic Processes, Tensor Networks, Math Theory of Probability.
- **Programs & Activities**: Road to Silicon Valley Program (Cohort 6), Quantitative Finance Club, Directed Research Reading Program, Physics Learning Theory Seminar, ESL Instructor.

GPU & Systems Research

Algorithmic Robotics and Control Lab, Rutgers Computer Science — New Brunswick, NJ

Research Assistant (Advisor: Prof. Jingjin Yu) — *Jan 2026 – Present*

- Implemented and validated CUDA versions of pRRTC and FCIT*, parallelizing tree/graph search and collision checking for high-dimensional robot motion planning across 500 benchmark problems.
- Developing a graph-based search policy for dual-arm task-and-motion planning in Isaac Sim, learning from successful and failed planner traces to rank symbolic action sequences and predict geometric feasibility.

Kwan Lab, Rutgers Cancer Institute of New Jersey — New Brunswick, NJ

Computational Biology Research Assistant (Advisor: Prof. Kelvin Kwan) — *Jan 2026 – Present*

- Debugged legacy TensorFlow/Keras dependencies to build a reproducible DeepSTARR inference environment on HPC for sequence-based prediction of enhancer activity.
- Developed an AlphaGenome in silico deletion screen spanning 12 sites, 3 deletion sizes, and 2 cell types (72 perturbation conditions) to quantify TAD-boundary sensitivity.
- Built a 3D chromatin polymer simulator using overdamped Langevin dynamics and multi-scale TAD detection; validated the pipeline with 70+ automated tests.

Rutgers Economics Labs — New Brunswick, NJ

President (formerly Research Director, Team Lead) — *Oct 2024 – Present*

- Architected RAIL, an automated research pipeline that collects, organizes, analyzes, and synthesizes economic data through data ontologies and coding-agent workflows.
- Lead quantitative research for NJDOL, NJDEP, NJBPU, Federal HUD, and the U.S. Census Bureau, applying econometric models in Python/R to large public datasets.

Software Engineering Experience

New York Life Insurance — New York, NY

Software Engineer Intern, TDAV Team — *May – Aug 2026*

- Built an AI text and voice agent on AWS for insurance claims workflows serving clients and 12,000 Customer Service Representatives, integrating enterprise services, knowledge, and forms.
- Consolidated business logic from 13,000 documents across SharePoint, Confluence, Jira, public sites, and the intranet into a knowledge platform for AI and human users.
- Developed cloud-hosted multi-agent workflows supporting AEM development, Pega code, document processing, and mainframe modernization.

Lykke San Francisco, CA

Founding Software Engineer *Sep 2025 – Present*

- Built a distributed course-wiki engine with tenant-isolated background jobs using AWS SQS/ECS, MongoDB leases, Redis rate limiting, and S3; implemented retrieval, streaming, and failure-handling paths.
- Engineered production data and RAG pipelines across MongoDB, Milvus, S3, and SSE, including source inference, citation events, token budgeting, and fallback paths for large course materials.
- Built ingestion workflows for Canvas LMS courses, assignments, announcements, notes, and uploaded documents, creating class-scoped knowledge bases for contextual chat and study generation.

Scarlet Sync New Brunswick, NJ

Founder *Jan 2025 – Present*

- Architected a Python data pipeline that reverse-engineers legacy university systems and indexes 2,500+ classes and 500+ degree programs for real-time querying by thousands of users.
- Built and shipped a production academic-planning platform with a streaming, tool-calling AI assistant across Web, ChatGPT, Claude, and iOS.

Samaritan Scout Cranford, NJ

Full Stack Engineer Intern *May 2023 – Aug 2025*

- Developed a Python extraction system over 100,000+ nonprofit websites using structured model outputs, reducing manual data entry by over 90%.
- Built a React, TypeScript, and PostgreSQL search product indexing 35,000+ volunteer opportunities with semantic matching and hallucination filters.

Selected Technical Projects

SLM Optimization & Post-Training | *PyTorch, HPC, Quantization, Pruning*

- Fine-tuned small language models on GSM8k and deployed them on Rutgers's Amarel HPC cluster, using quantization and structural pruning to improve inference throughput on constrained compute.

Knowledge Base & Agent Workflow Manager | *Python, MCP/CLI, Hybrid Retrieval*

- Built KRAIL, a local-first knowledge runtime for ontology-aware pages, source ledgers, claim records, verification runs, artifact lineage, and hybrid keyword-graph-vector retrieval.
- Implemented auditable agent infrastructure with local runner dispatch, dry-run work orders, session logs, and workflow validation.

Native macOS Computer-Use Agent | *Swift, MCP, Gemini Vision* (private code)

- Built a native desktop agent integrating vision models with macOS Accessibility APIs, AppleScript, and JXA; orchestrated specialized UI, shell, and application-research agents through MCP.

Technical Skills

Languages: C++, CUDA, Python, Java, JavaScript/TypeScript, SQL, Bash, R, Git | **Systems**: Distributed Systems, AWS (SQS, ECS, S3), Redis, MongoDB, PostgreSQL, HPC
ML & Data: PyTorch, TensorFlow/Keras, NumPy, Pandas, Transformers, Quantization, Inference, Automated Testing
Awards: 8090 AI Top Coder (5th), Rutgers Shark Tank (2nd and 3rd), ASA Fall Data Challenge (Top 3), NSF I-Corps